Rainfall Forecasting with Variational Autoencoders and LSTMs

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***Abstract*—In this paper we present a case study using a novel machine learning system for rainfall forecasting in a localized area over Colombia, South America. We explore a new forecasting approach inspired by established techniques used in computer vision and generative modeling to create a predictive model for precipitation maps. Using an ensemble made of a Variational Autoencoder and a stacked LSTM we were able to create a system which learns the spatial and temporal features of weather patterns in an integrated way, but also allows them to be measured and studied independently. Such a system introduces practical benefits in applications such as detection of rare weather patterns or analysis of anomalous events in addition to its regular forecasting capabilities.**

***Index Terms*—Weather Forecasting, Rainfall, LSTM, Variational Autoencoder, Deep Learning**

I. Introduction

Rainfall forecasting is a critical component of weather prediction and disaster management. In an era of climate change, accurate and adaptable rainfall forecasting has become increasingly important as it has a direct impact on the lives and well being of people around the world [1]–[6]. This paper presents a new system for rainfall forecasting that utilizes unsupervised feature learning to provide both accurate predictions and analysis capabilities which could be used to detect and study rare weather patterns. This system introduces a novel system for rainfall forecasting, leveraging a design based on the VAE-LSTM [7], [8] popularized in computer vision to better understand the impact of climate change on precipitation patterns. The objective of this study is to create and test a new, analysis friendly, and modular method for forecasting rainfall; which will benefit communities, governments, and businesses alike.

II. Background Information

A. Autoencoders and Feature Learning

Identifying features has long been a fundamental concept in machine learning, but in many cases this requires the data to have features identified by hand and labeled in all the training data. More recent developments in deep learning have led to the creation of several types of machine learning models that can identify and represent these features automatically from the patterns in the data itself with little or no labeling required. An autoencoder is one such model.

The autoencoder's origins are comparatively old and disputed, and many very similar concepts existed under different names. But it is commonly attributed to Yann LeCun's paper in 1987 [9]. An Autoencoder's purpose is to learn features in a dataset by training two neural networks together as one. The autoencoder is divided into symmetrical halves, which can each be used independently after training. When connected together they are shaped like an hourglass. See Figure 1 for a diagram of an autoencoder. The first network, named the encoder, learns to represent the input data in a smaller format. The second model, called the decoder, takes data in that smaller format as input and learns to recreate the original data.

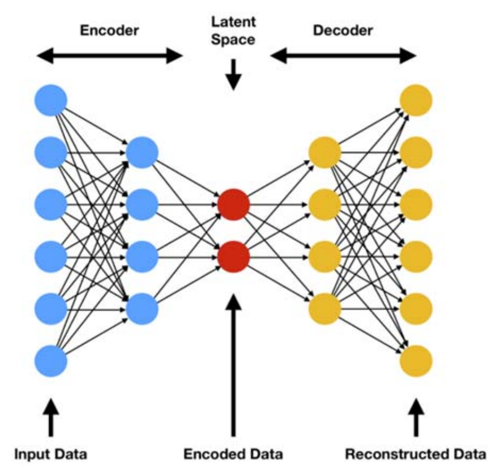


Fig. 1: Diagram of autoencoder architecture [10]

By doing this the model learns to encode features in the data automatically without requiring labeling. The vector which the encoder outputs is sometimes called a feature representation or latent representation. An autoencoder is trained by connecting the encoder's output to the decoder's input, and then training using backpropagation with the target output the same as the input data.

B. Variational Autoencoders

The autoencoder used in this project has been modified into a variational autoencoder [11], [12] which uses an additional loss term when training the encoder called Kullback Leibler Divergence. KL Divergence is a type of statistical distance, which is used to compare probability distributions. This change guides the autoencoder's latent space to take the form of a multivariate normal distribution.

In Aqeel Anwar's publication in Toward Data Science titled "Difference between AutoEncoder (AE) and Variational AutoEncoder (VAE)" [13] the author visualizes the quality that sets VAEs apart from regular autoencoders by graphing his VAE's latent space after it is trained to represent handwritten digits zero through nine. He sets the dimensionality of the latent vector to two so that it can be viewed on a two dimensional plot. Figure 2 shows the comparison.

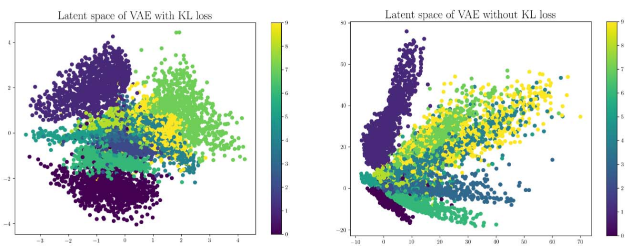


Fig. 2: Categorical distribution of the latent space of a Variational Autoencoder(Left) and an Autoencoder(Right) [13]. The color of the points indicate which class they belong to, while the x and y axes of the plot are the two dimensions of the point's latent representation.

Notice how the VAE's latent space is regularized and centered around zero, while the center and bounds of the AE's latent space is arbitrary. Also note the improved clustering behavior of the VAE with it's more distinctive class boundaries.

C. Long Short Term Memory Networks

In addition the the VAE's encoder and decoder, the final ensemble includes a third model whose purpose is to forecast the next feature vector in a sequence. To fulfill this purpose we experimented with a few types of recurrent neural network (RNN), but eventually settled on a model made up of stacked LSTMs.

Long Short Term Memory Networks [14]–[16], or LSTMs for short, are a type of RNN which were designed to handle long term dependencies in the input sequence better than their predecessors which have been successfully applied to many types of sequential data [17]–[19]. The forecaster model uses several LSTMs stacked five units deep. Stacking additional units has a similar effect to adding more layers to a feed forward network, making it able to learn more complex patterns at the cost of increased effect of the vanishing gradient problem.

III. Dataset

The data used in this work is a subset of the ERA5-Land dataset provided by The European Center for Medium Range Weather Forecasts(ECMWF) and the Copernicus Climate Change Service. ECMWF maintains some of the most sophisticated weather forecasting and analysis systems in the world, as well as a wide array of publicly available weather and climate datasets via the Climate Data Store [20].

The subset of ERA5-Land used in this project is geographically centered over Colombia, South America and is a reanalysis dataset comprised of hourly maps of daily accumulated rainfall beginning on January 1st, 1981. ERA-5 Land itself is much more comprehensive, with global coverage, older data dating back to 1950, and more than 50 types of measurements.

Before the data is used to train any Keras models it is stored in a NumPy array and normalized to the range 0-1. The data is then split and saved with the last year of data reserved as the test set.

IV. System Design

First the normalized training data is used to train a variational autoencoder (VAE) [11]. See Figure 3 for a comparison between a precipitation map from the test set, and the same precipitation map after being encoded and decoded by the VAE. After training the VAE, the encoder is separated from the decoder and they are saved. The entire dataset is then inferenced by the encoder to create a new dataset of feature-encoded vectors for each rainfall map. This dataset will be used for the other parts of the system which operate on feature representations of the rainfall maps rather than raw data.

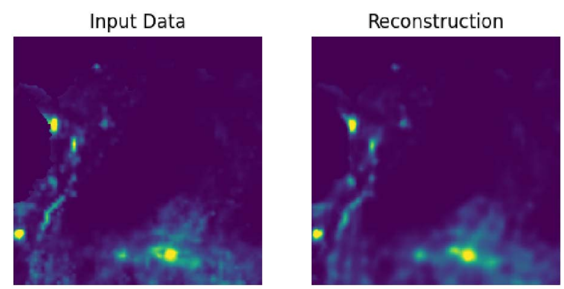


Fig. 3: VAE reconstruction of a rainfall map sampled from the test dataset

Once a satisfactory VAE is trained and the encoded dataset generated, the new dataset is used to train a stacked LSTM [14], [15] to take a sequence of feature vectors as input and return a prediction for the next feature vector in the sequence.

After the LSTM is trained, the models are ready to be inferenced as an ensemble. A sequence of precipitation maps is first passed to the VAE encoder, which is repeatedly applied to encode each map in the sequence to feature vectors. That vector sequence is passed to the stacked LSTM, which outputs a predicted next vector for the input sequence. That predicted vector is then passed as input to the VAE decoder, which constructs a rainfall map from it and that is the ensemble's output. See Figure 4 for a diagram describing the architecture.

When training the models, the VAE uses a combination of Kullback-Leibler divergence and mean squared error loss as

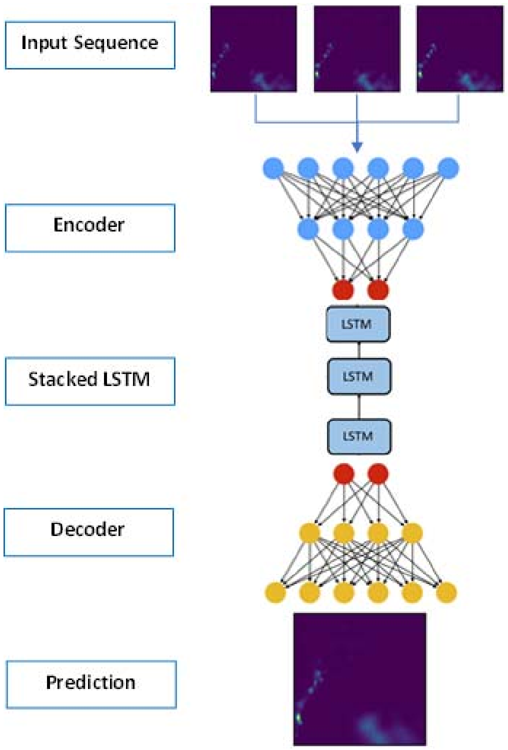


Fig. 4: Diagram of the complete ensemble

is typical for VAEs. The stacked LSTM was trained using euclidean distance loss, which was chosen experimentally out of several loss functions.

V. Experimental Results

The VAE-LSTM was able to learn the weather patterns very well, although some parts of the ensemble generalized to test data better than others. See Figure 8 for a sequence of predictions by system and actual values taken from the test set.

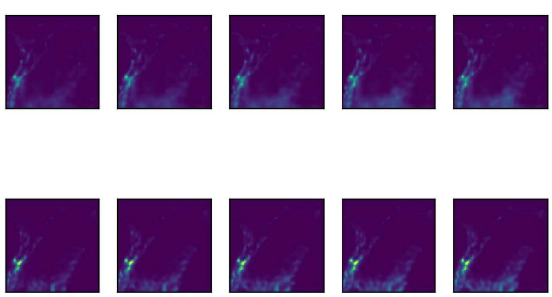
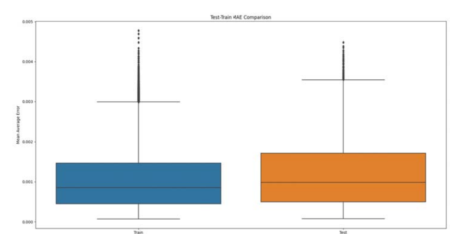


Fig. 5: Sample series of forecasts from the test set(top) and the true values(bottom)

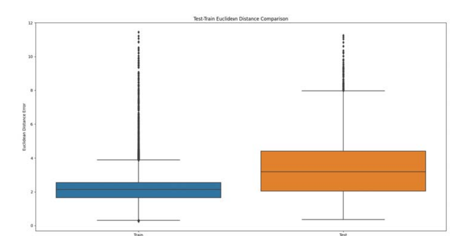
VAEs generate realistic results for the samples in their training set [21]. At all other points in the latent space outside of the training set, the reconstructed features appear as a blending of the closest samples in the latent space. The most prominent effect is that features will tend to blur together due to the VAE's assumption of independence of its latent variables.

This assumption causes the VAE to "distribute probability mass diffusely over the data space" [22] instead of concentrating it into spatially independent features. This behavior is an improvement over the original autoencoder, because it helps the VAE generate more realistic and cohesive output for latent points which lie outside of the training set. In our case it captures the overall shape of the spatial and temporal features of the data quite well, and this quality generalizes to the test set.

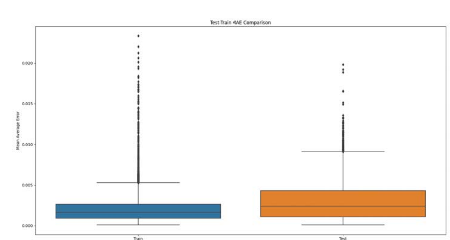
The VAE generalized to the test set much better than the stacked LSTM as shown in Figure 6. It is important to note that the divergence of test set and training set performance is what is comparable across plots, not the error values. From these error distributions we can see that the VAE performed very similarly at representing data from the test and training set, while the stacked LSTM was significantly better at predicting feature vectors for the training set than the test set.



(a) VAE Error



(b) Stacked LSTM Error



(c) Whole-Ensemble Forecast Error

Fig. 6: Boxplots of train(blue) and test(orange) error distribution for models in the ensemble.

By analyzing the latent representations we were also able to visualize some trends at the feature level. While individual features are very noisy and difficult to make sense of, Fourier analysis showed that each feature has some frequencies which are much more significant than others. When we used the significant frequencies to approximate the test data the simple methods we employed did not generalize to the test set but further analysis is warranted. Some weather patterns lasting many hours were able to be represented by a repeated linear vector translation through the latent space, whereas others were not.

By summing up the hourly change in every learned feature, it is possible to see patterns in the total rate of change of all features. It seems to have both repeating daily patterns and longer term ones. See Figure 7 for an example of these daily patterns. The ”change” value is determined by calculating the distance between the feature representation of a rainfall map and the next one in the sequence. The large spike every 24 hours is the accumulated values resetting, and between the spikes you can see a common daily pattern as well as longer multi-day trends.

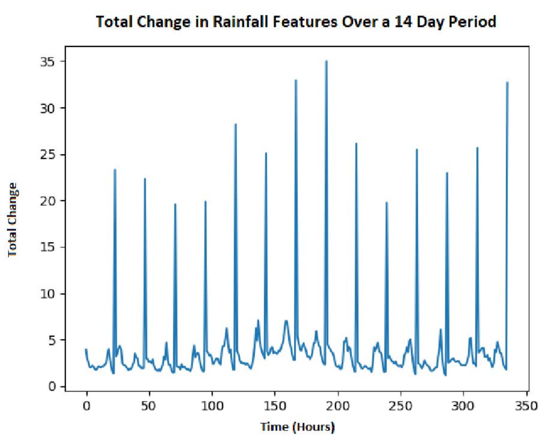


Fig. 7: Total change in rainfall map features over a 14 day period

VI. Conclusion and Future Work

Overall the model was successful at what we set out to do. The architecture has already demonstrated potential for many applications. The general paradigm of feature learning is proving to be very applicable to weather and climate systems, and we look forward to expanding the system’s capabilities for both prediction and analysis in future research.

We have identified several directions in which we would like to expand this research in the future. Measuring the system’s performance with a benchmark dataset is a must. Adoption of a GAN-based architecture [23], [24] could lead to higher quality modeling of spatial features than the VAE can provide. Replacing the stacked LSTM with a seq2seq LSTM encoder-decoder [25], [26], or a transformer [27], would allow for predictions of sequences of output and may lead to improved generalization over the current LSTM.

Alterations to the VAE to control the shape of the latent space could make the data easier for the forecasting model to predict. Learned-feature analysis of weather and climate data should also be expanded upon as it may offer useful insights into patterns and trends in the historical data as well as means of automatically identifying identifying anomalies or interesting trends.

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Contains modified Copernicus Climate Change Service Information [2022]

References

[1] A. Mishra, C. Siderius, K. Aberson, M. van der Ploeg, and J. Froebrich, “Short-term rainfall forecasts as a soft adaptation to climate change in irrigation management in north-east india,” *Agricultural Water Management*, vol. 127, p. 97–106, 2013.

[2] J. I. Kuwajima, F. M. Fan, D. Schwanenberg, A. Assis Dos Reis, A. Niemann, and F. F. Mauad, “Climate change, water-related disasters, flood control and rainfall forecasting: A case study of the são francisco river, brazil,” *Geological Society, London, Special Publications*, vol. 488, no. 1, p. 259–276, 2019.

[3] E. J. Kendon, N. M. Roberts, H. J. Fowler, M. J. Roberts, S. C. Chan, and C. A. Senior, “Heavier summer downpours with climate change revealed by weather forecast resolution model,” *Nature Climate Change*, vol. 4, no. 7, p. 570–576, 2014.

[4] D. López-Carr, N. G. Pricope, J. E. Aukema, M. M. Jankowska, C. Funk, G. Husak, and J. Michaelsen, “A spatial analysis of population dynamics and climate change in africa: Potential vulnerability hot spots emerge where precipitation declines and demographic pressures coincide,” *Population and Environment*, vol. 35, no. 3, p. 323–339, 2014.

[5] D. Lopez-Carr, K. M. Mwenda, N. G. Pricope, P. C. Kyriakidis, M. M. Jankowska, J. Weeks, C. Funk, G. Husak, and J. Michaelsen, “Climate-related child undernutrition in the lake victoria basin: An integrated spatial analysis of health surveys, ndvi, and precipitation data,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 9, no. 6, p. 2830–2835, 2016.

[6] J. Edwards, “Development and comparison of machine and deep learning models for the prediction of land degradation,” Ph.D. dissertation, University of North Carolina Wilmington, 2022.

[7] A. J. Lew and M. J. Buehler, “Encoding and exploring latent design space of optimal material structures via a vae-lstm model,” *Forces in Mechanics*, vol. 5, p. 100054, 2021.

[8] Z. Niu, K. Yu, and X. Wu, “Lstm-based vae-gan for time-series anomaly detection,” *Sensors*, vol. 20, no. 13, p. 3738, 2020.

[9] Y. Le Cun and F. Fogelman-Soulié, “Modèles connexionnistes de l’apprentissage,” *Intellectica. Revue de l’Association pour la Recherche Cognitive*, vol. 2, no. 1, p. 114–143, 1987.

[10] S. Flores, “Variational autoencoders are beautiful,” Apr 2019. [Online]. Available: https://www.compthree.com/blog/autoencoder/

[11] D. P. Kingma and M. Welling, “Auto-encoding variational bayes,” 2013. [Online]. Available: https://arxiv.org/abs/1312.6114

[12] ——, “Auto-encoding variational bayes,” May 2014. [Online]. Available: https://arxiv.org/abs/1312.6114

[13] A. Anwar, “Difference between autoencoder (ae) and variational autoencoder (vae),” Nov 2021. [Online]. Available: https://towardsdatascience.com/difference-between-autoencoder-ae-and-variational-autoencoder-vae-ed7be1c038f2

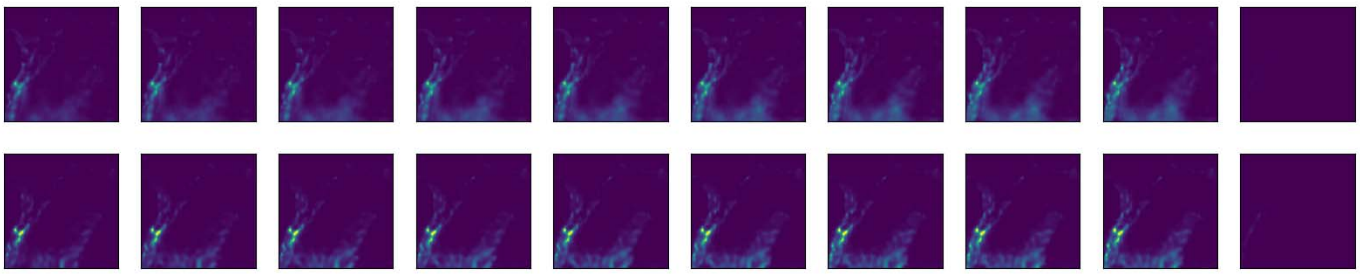
[14] A. Graves, A.-r. Mohamed, and G. Hinton, “Speech recognition with deep recurrent neural networks,” 2013. [Online]. Available: https://arxiv.org/abs/1303.5778

[15] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, p. 1735–1780, nov 1997. [Online]. Available: https://doi.org/10.1162/neco.1997.9.8.1735
[16] H. Sak, A. Senior, and F. Beaufays, "Long short-term memory based recurrent neural network architectures for large vocabulary speech recognition," Feb 2014. [Online]. Available: https://arxiv.org/abs/1402.1128
[17] E. Kurpiewski, I. Samokhvalov, B. Layne, L. Rasaq, G. Dogan, and M. Heijnen, "Classification of activities and falls within a multimodal dataset," *2021 IEEE International Conference on Machine Learning and Applied Network Technologies (ICMLANT)*, 2021.
[18] G. Dogan, M. Ford, and S. James, "Predicting ocean-wave conditions using buoy data supplied to a hybrid rnn-lstm neural network and machine learning models," *2021 IEEE International Conference on Machine Learning and Applied Network Technologies (ICMLANT)*, 2021.
[19] G. Dogan, "Advances in human activity recognition," *Computer*, vol. 54, no. 5, p. 4–6, 2021.
[20] Muñoz Sabater J., "Era5-land hourly data from 1981 to present," 2019 (Accessed Jan-2021).
[21] Jonnalagadda Venkata Krishna, "Sparse stacked and variational autoencoder," *Medium*, Dec 2018. [Online]. Available: https://medium.com/@venkatakrishna.jonnalagadda/sparse-stacked-and-variational-autoencoder-efe5bfe73b64
[22] V. Dumoulin, I. Belghazi, B. Poole, O. Mastropietro, A. Lamb, M. Arjovsky, and A. Courville, "Adversarially learned inference," 2016. [Online]. Available: https://arxiv.org/abs/1606.00704
[23] S. Ravuri, K. Lenc, M. Willson, D. Kangin, R. Lam, P. Mirowski, M. Fitzsimons, M. Athanassiadou, S. Kashem, S. Madge, R. Prudden, A. Mandhane, A. Clark, A. Brock, K. Simonyan, R. Hadsell, N. Robinson, E. Clancy, A. Arribas, and S. Mohamed, "Skilful precipitation nowcasting using deep generative models of radar," *Nature*, vol. 597, 2021.
[24] H.-B. Liu and I. Lee, "Mpl-gan: Toward realistic meteorological predictive learning using conditional gan," *IEEE Access*, vol. 8, p. 93179–93186, 2020.
[25] I. Sutskever, O. Vinyals, and Q. V. Le, "Sequence to sequence learning with neural networks," 2014. [Online]. Available: https://arxiv.org/abs/1409.3215
[26] K. Cho, B. van Merrienboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using rnn encoder-decoder for statistical machine translation," 2014. [Online]. Available: https://arxiv.org/abs/1406.1078
[27] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," 2017. [Online]. Available: https://arxiv.org/abs/1706.03762

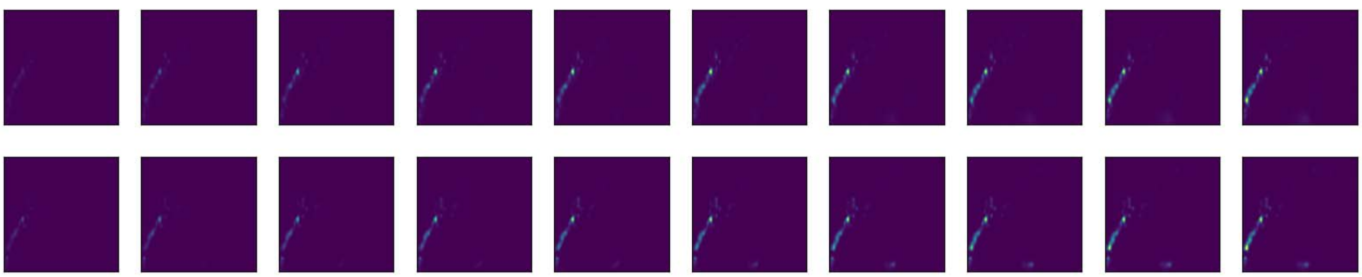
APPENDIX

The following page contains additional results sampled from the test set. The full set of output can also be furnished upon email request.

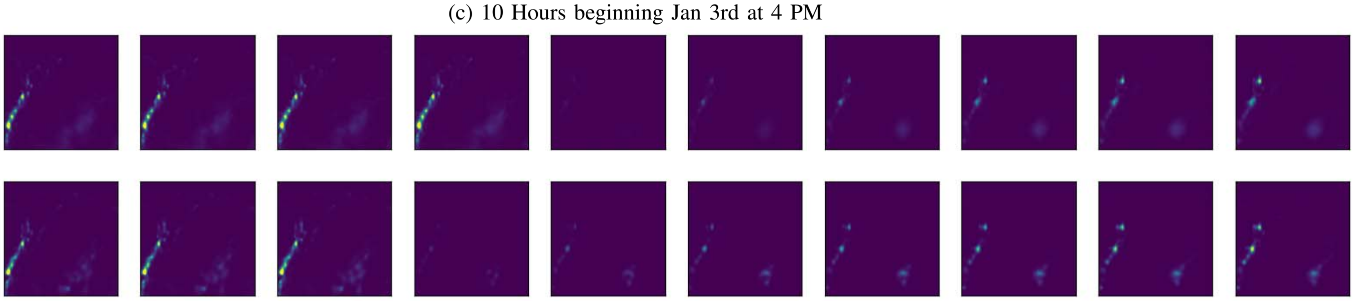
(a) 10 Hours beginning Jan 2nd at 8 PM



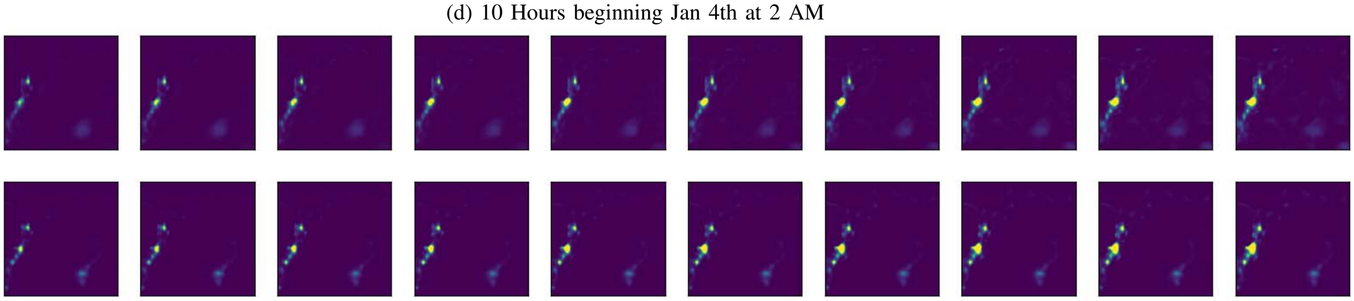
(b) 10 Hours beginning Jan 3rd at 6 AM



(c) 10 Hours beginning Jan 3rd at 4 PM



(d) 10 Hours beginning Jan 4th at 2 AM



(e) 10 Hours beginning Jan 4th at 12 PM

Fig. 8: Larger series of predictions(top) and observations(bottom) from the test set.